Variable-Thrust, Variable-Mass Rockets: Near-Earth Flight and Control

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01 Problem & physical model

Orbit insertion requires both altitude and velocity. Near-Earth orbits enable observation, space experiments and subsequent missions. Reaching altitude without sufficient transverse speed yields a suborbital return.

Vertical ascent and ideal propulsion

For expelled mass $-\mathrm{d}m > 0$ and relative exhaust speed v_e (also denoted by c below), momentum balance gives

$$m\,\mathrm{d}v = -v_e\,\mathrm{d}m + F_{\text{ext}}\,\mathrm{d}t.$$

Neglecting external forces and integrating yields the Tsiolkovsky equation:

$$\boxed{\Delta v_{\text{ideal}} = c\ln(m_0/m_f)}.$$

Impulse is $I_T = \int T\,\mathrm{d}t = v_e(m_0 - m_f)$; velocity gain integrates T/m. Starting from rest with $r \geq r_0$, no random forcing and dissipative drag, ideal propulsion and orbital energy imply

$$\frac{m_0}{m_f} \geq \exp\left[\frac{\sqrt{2\mu/r_0 - \mu/r_*}}{v_e}\right].$$

The necessary mass ratio is at least **6.245** in the 600 km case; this bound is not sufficient for insertion.

Earth-centered planar polar dynamics

Define $u = \dot{r}$, $w = r\dot{\theta}$, $V = \sqrt{u^2 + w^2}$, propellant flow $q = -\dot{m} \geq 0$ and thrust $T = cq$.

$$\begin{aligned}
\dot{r} &= u, \qquad \dot{\theta} = w/r, \qquad \dot{m} = -q,\\
\dot{u} &= \frac{w^2}{r} + \frac{cq\cos\alpha}{m} - \frac{\mu}{r^2} - \frac{kVu}{m} + a_{\varepsilon,r},\\
\dot{w} &= -\frac{uw}{r} + \frac{cq\sin\alpha}{m} - \frac{kVw}{m} + a_{\varepsilon,\theta}.
\end{aligned}$$

Here $k = \rho C_D A/2$ and $\rho = \rho_0 e^{-(r-R_E)/H}$. Thrust follows $\alpha = \operatorname{atan2}(w,u) + \delta$ after a short initial vertical hold.

$$a_{\varepsilon,j} = \sigma_{0,j}(\rho/\rho_{\text{ref}})\eta_j(t).$$

Gaussian forcing is continuously interpolated with finite temporal correlation, not ideal white noise. The planar point-mass model omits Earth rotation, lift and attitude actuator dynamics.

Physical settings and capability

Initial / dry mass	60 000/1 200 kg
Effective exhaust speed	4 500 m/s
Drag coefficient / area	0.35/2.0 m^2
Density / scale height	1.225 kg/m^3 / 8.5 km
Disturbance at reference density	0.1 m/s^2

Circular speed is about 7.562 km/s; the energy-based ideal velocity bound is 8.243 km/s. The achieved mass ratio is 9.636, above the necessary bound, reflecting finite thrust, steering and external-force losses.

Numerical integration links control to flight

For $\dot{y} = f(t,y)$, classical RK4 evaluates four slopes:

$$y_{n+1} = y_n + \frac{\Delta t}{6}(K_1 + 2K_2 + 2K_3 + K_4).$$

Batched GPU RK4 can also accelerate simulation and substantially reduce computation time.

The mass-ratio lower bound from Tsiolkovsky's equation guides initialization and feasibility screening for ODE-based trajectory optimization.

02 Planning & feedback control

Trajectory planning: PSO search with SQP refinement

A suborbital burn, coast and circularization burn structure the search over thrust-angle nodes, flow rates and phase durations. PSO combines personal and neighborhood experience; SQP then refines terminal feasibility.

$$r_* = R_E + 600\,\text{km}, \qquad v_c = \sqrt{\mu/r_*},$$

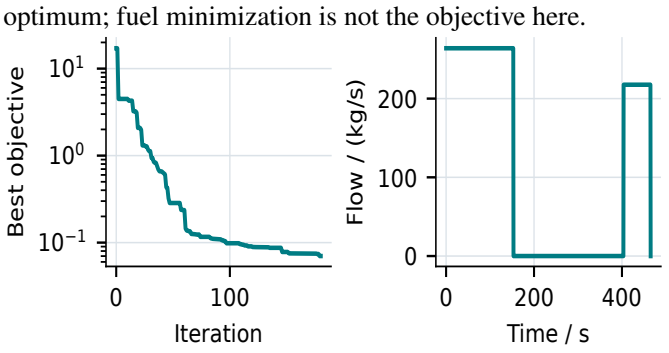
$$J = \left(\frac{r_f - r_*}{s_h}\right)^2 + \left(\frac{u_f}{s_u}\right)^2 + \left(\frac{w_f - v_c}{s_w}\right)^2.$$

How PSO works

In normalized design space, particle position and search velocity are ξ_i, v_i; personal and neighborhood bests are p_i, g_i:

$$\begin{aligned}
v_i^{k+1} &= \omega v_i^k + c_1 r_1 \odot (p_i - \xi_i^k)\\
&\quad + c_2 r_2 \odot (g_i - \xi_i^k),\\
\xi_i^{k+1} &= \xi_i^k + v_i^{k+1}.
\end{aligned}$$

Random exploration and shared experience reduce terminal error. Bounds, impact and propellant-budget violations are constrained or penalized. PSO does not guarantee a global optimum; fuel minimization is not the objective here.



Left: PSO objective history. Right: final reference flow. Burn/coast/burn durations are approximately 153.3 / 250.0 / 62.0 s. Final accuracy reflects both PSO and local refinement.

MPC: measure, predict, solve, apply

Track $x^{\text{ref}} = (r, \theta, u, w, m)^{\mathsf{T}}$ using thrust-angle and flow corrections. Linearizing the discrete dynamics along the reference gives

$$e_{k+1} = A_k e_k + B_k v_k + d_k.$$

Receding-horizon cost and constraints

$$\begin{aligned}
J_{\text{MPC}} = &\sum_{j=1}^{N} \|e_{k+j|k}\|_{Q_j}^2 + \sum_{j=0}^{N-1} \|v_{k+j|k}\|_R^2\\
&+ \sum_{j=0}^{N-1} \|\Delta v_{k+j|k}\|_{R_\Delta}^2 + \lambda\|\epsilon\|^2.
\end{aligned}$$

The tracking error e and control correction v are normalized by physical scales. Ground clearance is softened; dry-mass and input limits are enforced as hard constraints.

Quadratic penalties on predicted tracking errors, control effort, control changes and slack variables are condensed into a convex quadratic program:

$$\min_{\zeta}\ \tfrac{1}{2}\zeta^{\mathsf{T}} H\zeta + f^{\mathsf{T}}\zeta, \qquad G\zeta \leq b.$$

A **convex interior-point algorithm** solves the QP with angle, flow, dry-mass and ground constraints. Apply only the first correction, then repeat with fresh state feedback; retain the two-burn schedule.

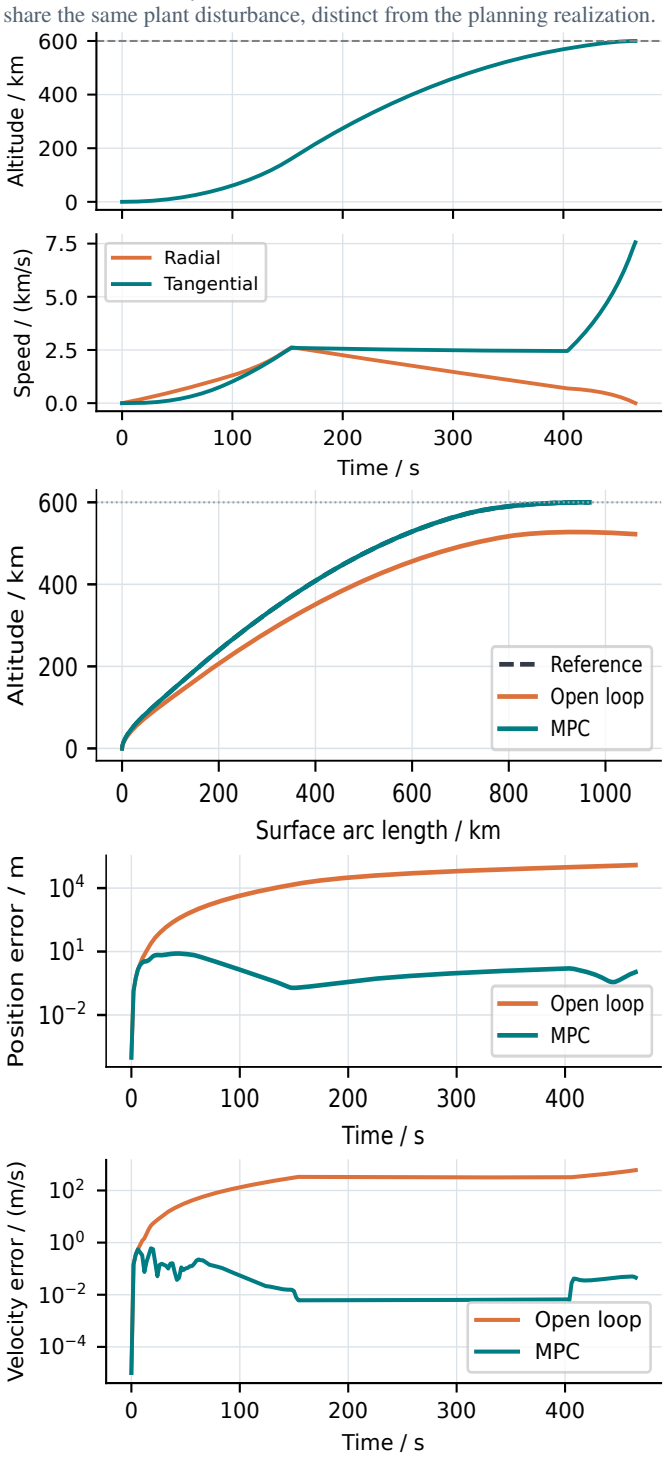
Why is the QP convex?

After state elimination, $H = 2(\mathcal{B}^{\mathsf{T}}\bar{Q}\mathcal{B} + \bar{R} + D^{\mathsf{T}}\bar{R}_\Delta D)$. Positive-definite input weighting makes H positive definite. The interior-point solver follows perturbed KKT conditions as the barrier parameter decreases.

Offline planning supplies a feasible reference; online MPC corrects disturbances in flight.

03 600 km case & findings

$m_0 = 60\,000$ kg, $m_{\text{dry}} = 1\,200$ kg, $c = 4\,500$ m/s. Open and closed loops share the same plant disturbance, distinct from the planning realization.



MPC nearly overlaps the reference trajectory. Position error combines radial and surface-arc errors; zero values are floored only for logarithmic plotting.

Metric	Open loop	MPC
Position RMS / m	5.98×10^4	2.606
Velocity RMS / (m/s)	298.084	0.0992
Cutoff altitude error / m	−77 611.6	1.191
Cutoff radial speed / (m/s)	−557.983	0.0231
Tangential speed error / (m/s)	233.256	0.0559

RMS is relative to the reference; cutoff errors are relative to the target circular orbit.

Flight phases and online computation

The first burn builds ascent and transverse motion; coasting gains altitude; the second burn raises transverse speed and reduces radial speed. Reference cutoff residuals are 0.265 m, −0.0212 m/s and 0.0463 m/s. MPC updates every 2 s with a 30 s horizon. All 230 QPs succeeded; mean solve time was 1.793 ms.

Feedback suppresses error accumulation and brings cutoff close to circular-orbit conditions.

References: NASA Glenn, Ideal Rocket Equation; Kennedy & Eberhart, Particle Swarm Optimization, 1995; Dormand & Prince, JCAM, 1980; Mayne et al., Automatica, 2000; Nocedal & Wright, Numerical Optimization, 2006. Data: project experiment snapshot, 20 Oct 2026.